

Emilio Hernandez-Martinez

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EDUCATION

University of Michigan College of Engineering

August 2025 — May 2029

Bachelor's of Science in Engineering - Major in Mechanical Engineering / GPA: 3.94

Ann Arbor, MI

- **Relevant Coursework:** Introduction to Engineering, Introduction to Computers and Programming, General Physics I, Introduction to Academic Research, & Introduction to Differential Equations
- **Additional Activities:** Michigan Research and Discovery Scholars (MRADS), FIRST Alumni & Mentors Network at Michigan (FAMNM)

Forest Hills Northern High School

August 2021 — May 2025

High School Diploma / GPA: 4.50

Grand Rapids, MI

- **Relevant Coursework:** AP Calculus BC, AP Computer Science A, AP Computer Science Principles, AP Physics 1, AP Physics 2, & Multivariable and Vector Calculus
- **Awards:** Congressional Medal of Merit Recipient, National Merit Scholar, & Forest Hills Business Association Scholarship Recipient

EXPERIENCE

University of Michigan Quantum Engineering Laboratory

September 2025 — Present

Undergraduate Researcher

Ann Arbor, MI

- Design, fabricate, and test in-house optical hardware, including a 3D-printed fiber polarization controller.
- Perform finite-element simulations to study the physical stability of structures inside of a dilution refrigerator.
- Develop MATLAB tools to analyze simulated HBAR mode behavior and optimize resonator geometries.

FIRST Robotics Competition Team 3357 COMETS

July 2021 — July 2025

Mechanical Lead, Pit Crew Member, and Robot Operator

Grand Rapids, MI

- Managed the design and manufacturing of FRC robots, specializing in CAD and CNC operation.
- Mentored younger members in CAD software, machine shop tools, and basic robot assembly.
- Analyzed scouting data regarding competition performance and planned match strategies with other teams.

National Honor Society

August 2024 — May 2025

Tutoring Coordinator

Grand Rapids, MI

- Connected students in need of tutoring with capable NHS members and registered earned volunteering hours.
- Held open tutoring sessions twice a week and presented volunteer activities to NHS members in monthly meetings.

Grand Rapids Public Library

September 2023 — April 2025

MINDSTORMS Saturdays Volunteer

Grand Rapids, MI

- Organized an open workshop once a month throughout the school year at a local library.
- Taught elementary-age students STEM fundamentals through building and programming LEGO MINDSTORMS.

Barnes & Noble College Book Store

July 2024 — September 2024

Campus Retail Associate

Grand Rapids, MI

- Assisted college students at Grand Rapids Community College with finding and buying textbooks.
- Received bulk textbook shipments, packed and processed online orders, and organized product shelves.

SKILLS

- **Programming Languages:** Python, HTML, CSS, Java, JavaScript, MATLAB & Simulink
- **Software Applications:** SolidWorks, AutoCAD, Autodesk Fusion 360, Onshape, COMSOL, & Microsoft 365
- **Additional Skills:** Spanish (Native Speaker with Michigan Seal of Bilingualism)